Date: August 8, 2026

To:
The Editorial Board
Journal of Computer Science & Technology (JCS&T)

Subject: Submission of Manuscript entitled "Delta BPTT: Efficient Add-Only Backpropagation Through Time for Spiking Sequence Representations"

Dear Editors,

I am pleased to submit our original research manuscript entitled **"Delta BPTT: Efficient Add-Only Backpropagation Through Time for Spiking Sequence Representations"** for consideration for publication in the *Journal of Computer Science & Technology (JCS&T).*

1. Why this manuscript should be published in JCS&T:

Spiking Neural Networks (SNNs) are gaining rapid interest as a low-power alternative to conventional Deep Neural Networks for edge AI and neuromorphic computing. However, while forward passes in SNNs achieve high event-driven sparsity, the backward pass during Backpropagation Through Time (BPTT) traditionally re-introduces dense floating-point Multiply-Accumulate (MAC) operations, creating a significant computational bottleneck during training.

In this manuscript, we present **Delta BPTT**, a novel Add-Only gradient routing algorithm that exploits forward spike events as binary gates to replace dense weight-gradient accumulation with sparse, conditional scalar additions. Our evaluation demonstrates that Delta BPTT eliminates dense MACs in weight updates (67.2% sparsity matching forward pass) and achieves 2.44× faster training while maintaining strong empirical gradient alignment ($\cos(\nabla W_D, \nabla W_B) = 0.9982$) with Standard BPTT. Furthermore, we demonstrate linear $\mathcal{O}(L)$ sequence scaling over quadratic $\mathcal{O}(L^2)$ spatial attention. We believe these findings on green, energy-efficient AI align strongly with the computational algorithms and emerging technologies scope of JCS&T.

2. Declarations and Policy Statements:

- **Originality Confirmation:** We confirm that this manuscript is an original work and is not a reworked version of a paper that has been published or submitted for publication elsewhere, nor does it contain preliminary data.
- **Policy Compliance:** There are no issues relating to journal policies. All co-authors have reviewed and approved the manuscript for submission.
- **Competing Interests:** The author declares no competing financial or personal interests.

3. Suggested Potential Peer Reviewers:

In accordance with JCS&T guidelines, we suggest the following 3 potential peer reviewers who possess relevant expertise in Spiking Neural Networks, BPTT, and Neuromorphic Computing, and have no conflict of interest:

1. **Prof. Kaushik Roy**
   Affiliation: Department of Electrical and Computer Engineering, Purdue University, USA.
   Institutional Email: `kaushik@purdue.edu`

ORCID: `https://orcid.org/0000-0002-6451-8453`
Expertise: Neuromorphic Computing, Spiking Neural Networks, Energy-Efficient AI Hardware.

2. **Prof. Emre Neftci**
   Affiliation: Institute for Neuromorphic Software (PGI-15), Forschungszentrum Jülich & RWTH Aachen University, Germany.
   Institutional Email: `e.neftci@fz-juelich.de`
   ORCID: `https://orcid.org/0000-0002-3642-4523`
   Expertise: Surrogate Gradient Learning, Spatiotemporal BPTT, Spiking Neural Networks.

3. **Prof. Giacomo Indiveri**
   Affiliation: Institute of Neuroinformatics, University of Zurich and ETH Zurich, Switzerland.
   Institutional Email: `giacomo@ini.uzh.ch`
   ORCID: `https://orcid.org/0000-0002-7109-1689`
   Expertise: Neuromorphic Cognitive Systems, Spiking Neural Network Learning Rules.

Thank you very much for your time and consideration of our manuscript. We look forward to hearing from you.

Sincerely,

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